

Autonomous Credit Intelligence: An Event-Driven, Self-Healing Multi-Agent Framework for Real-Time B2B Risk Assessment

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Abstract. Batch reviews of B2B accounts receivable often miss intraday credit deterioration because payment behaviour, financial fundamentals, and litigation signals are evaluated in disconnected, periodic cycles. This paper presents ACIS-X (Autonomous Credit Intelligence System, Extended), an event-driven multi-agent framework on Apache Kafka that continuously maintains a live per-customer risk score. Specialised agents coordinate exclusively through Kafka events across five core processing layers, an autonomous model governance plane, and a cross-cutting self-healing plane. The pipeline features native Kafka Exactly-Once Semantics (EOS), three-tier signal fusion, SHAP audit persistence, statistical drift tracking via Population Stability Index (PSI), online challenger model benchmarking, and heartbeat-driven recovery from committed offsets. To evaluate the architecture under diverse macroeconomic stress cases, we utilize a jump-diffusion synthetic stress test harness. The evaluation demonstrates real-time streaming throughput, horizontal consumer group scalability, autonomous fault detection and recovery dispatch with restart-storm prevention, and the contribution of behavioural refinement and multi-signal fusion under controlled synthetic evaluation.

Keywords: Multi-agent systems · Event-driven architecture · Credit risk intelligence · Apache Kafka · Exactly-Once Semantics · Self-healing systems

1 Introduction

Management of enterprise credit risks calls for constant evaluation of payment status, financial position, and potential litigations. Daily or weekly batch cycles delay action when risk rises within a day. Event streaming platforms such as Apache Kafka [1] and emerging multi-agent designs [2,11] make it practical to maintain a live credit view instead of an overnight refresh. However, while recent deliberative LLM-based multi-agent systems [2,12] offer qualitative reasoning, their multi-second latency and non-deterministic outputs make them unsuitable

for high-throughput, sub-50 ms financial streaming pipelines subject to strict operational risk controls.

This work introduces ACIS-X (Autonomous Credit Intelligence System, Extended), a Kafka-based reactive, event-driven multi-agent framework (ED-MAS) grounded in the Actor model for real-time B2B credit intelligence. It targets three gaps in prior practice: (1) latency of batch monitoring; (2) siloed behavioural, financial, and litigation signals; and (3) fragile pipelines without autonomous recovery and drift governance.

The contributions are: a layered Kafka-only reactive agent architecture with a cross-cutting self-healing plane and autonomous model governance; end-to-end Kafka Exactly-Once Semantics (EOS); a three-tier fusion path combining behavioural metrics, external financials, and litigation into one score in $[0, 1]$ backed by regulatory policy guardrails; an autonomous governance agent supporting model-risk governance principles aligned with SR 11-7 via PSI drift tracking and challenger model benchmarking; heartbeat monitoring with deterministic restart from committed offsets; and a comprehensive test suite measuring latency, deduplication, scaling, predictive quality, weight sensitivity, and fault recovery under a jump-diffusion synthetic stress test harness.

2 Related Work

Classical credit scoring such as Altman’s Z-score [3] and recent explainable ML for enterprise credit [4,5] remain tied to offline or periodic feature tables, making them a poor fit for intraday monitoring even when SHAP improves transparency [6]. Ensemble and hybrid learners raise accuracy in published studies [5], yet refresh on batch folds rather than continuous invoice streams. In parallel, Kafka-based event sourcing and Kafka–Flink pipelines have moved financial fraud analytics onto high-throughput streams [7,8], proving sub-second paths are achievable, though typically scoring transactions rather than a standing B2B credit profile fusing fundamentals and litigation.

Multi-agent credit architectures are emerging: MASCA [2], Kubam’s agentic framework [11], and CreditAgent [12] explore LLM reasoning and multi-agent reinforcement learning (MARL) for credit appraisal. However, these systems focus on deliberative, conversational analysis with multi-second response times, rather than high-throughput event streaming choreography. Self-healing ideas trace back to autonomic MAPE-K control [9]; recent streaming microservices add circuit breakers and replay patterns [10], but often restart entire containers without fine-grained, agent-local heartbeat timeouts.

Table 1 contrasts the operational and architectural characteristics of prior paradigms against ACIS-X.

As shown in Table 1, ACIS-X uniquely occupies the intersection of streaming credit fusion, deterministic compliance, autonomous governance, and fine-grained agent recovery: specialized reactive agents exchange only Kafka events, fuse three signal families online, and resume autonomously from the last committed offset.

Table 1. Architectural positioning across enterprise credit risk paradigms

Dimension	Batch XAI [4]	Stream Risk [8]	Cognitive MAS [2,11]	ACIS-X (Ours)
Latency Profile	Hours – Days	Sub-second (<1 s)	Multi-second (>2 s)	Sub-50 ms (P95 16 ms)
Signal Scope	Financial ratios	Single transaction stream	Text / financial reports	3-tier (AR + Fin + Legal)
Execution Model	Periodic batch scripts	Stream processing	Deliberative LLM / MARL	Reactive event-driven MAS
Atomicity Guarantee	DB ACID transactions	At-least-once / EOS	None (stateless LLM)	Native Kafka EOS + PK
Fault Recovery	Cold process restart	Node failover	None / prompt retry	Heartbeat + offset resume
Drift Governance	Offline re-training	External metric dashboard	Manual prompt tuning	Streaming PSI + Challenger

3 System Architecture

ACIS-X maintains a live risk score for each customer c_i :

$$\mathcal{R}(c_i, t) = \mathcal{F}(\mathcal{B}(c_i, t), \mathcal{E}(c_i, t), \mathcal{L}(c_i, t)) \in [0, 1], \quad (1)$$

where $\mathcal{B}$ is behavioural history, $\mathcal{E}$ external financial indicators, and $\mathcal{L}$ litigation exposure. Design targets are sub-second processing, usable scores under partial signals, zero silent loss of committed work across restart, and idempotent duplicate handling.

Figure 1 depicts the runtime architecture. Layers 1–4 sit above a shared Apache Kafka event bus; Layer 5 (persistence), Layer 6 (model governance), and the system plane sit below. Agents connect exclusively through Kafka: Layer 1 builds customers, invoices, payments, and behavioural metrics; Layer 2 enriches with financial APIs and litigation, then aggregates; Layer 3 predicts payment risk and refines the composite score; Layer 4 applies policy and collections actions; Layer 5 persists state; Layer 6 monitors drift and benchmarks challenger models; the system plane manages time ticks, monitoring, self-healing, DLQ watching, and runtime placement.

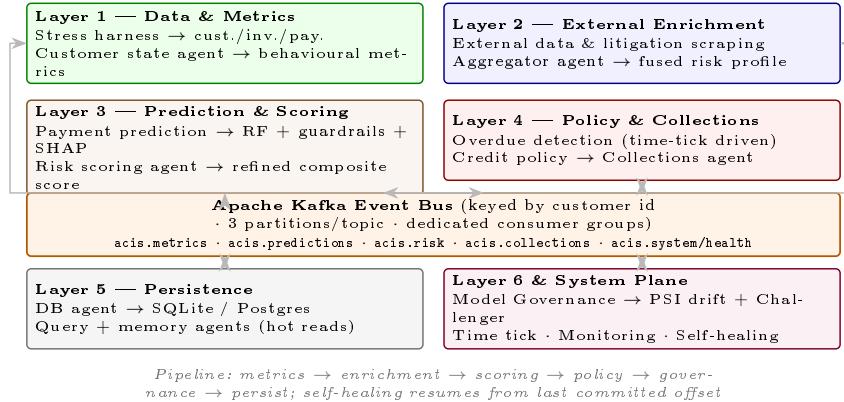


Fig. 1. ACIS-X layered architecture. Specialized reactive agents communicate exclusively through Kafka; the system plane manages health, discovery, and recovery.

3.1 Agent Inventory and Topic Wiring

Table 2 catalogues every agent across the processing layers, governance plane, and system plane, summarizing its layer, topic subscriptions, publications, and operational role.

Table 2. ACIS-X Unified Agent Inventory: Layer, topic routing, and operational role

Agent	L	Subscribes To	Publishes To	Role / Operational Rationale
ScenarioGenerator	1	acis.commands	cust., inv., pay.	Emits reproducible synthetic and jump-diffusion macro-stress streams.
CustomerState	1	invoices, payments	acis.metrics	Rebuilds behavioural metrics (OTR, delay, utilisation).
ExternalData	2	acis.metrics	acis.metrics	Enriches and caches financial ratios (D/E, ROE, PE).
LitigationScraping	2	acis.metrics	acis.metrics	Quantifies legal exposure from court/news feeds.
Aggregator	2	acis.metrics	acis.risk	Fuses Tier 2 financial and litigation signals (0.6/0.4).
PaymentPrediction	3	acis.metrics	acis.predictions	Random Forest scoring + guardrails + SHAP attributions.
RiskScoring	3	pred., cust., met., risk	acis.risk	Refines base score with behavioural & temporal velocity.
OverdueDetection	4	acis.time	acis.invoices	Time-tick driven detection of delinquency freshness.
CreditPolicy	4	acis.risk	acis.collections	Maps risk tiers to operational control intents.
Collections	4	acis.collections	acis.collections	Executes idempotent collection actions without duplicate storms.
DBAgent	5	ALL topics	SQLite / Postgres	Persists events for state recovery across process restarts.
Query / Memory	5	(synchronous)	(in-memory cache)	Serves hot in-memory reads with SQLite/Postgres fallback.
ModelGovernance	6	pred., metrics	acis.system, health	Tracks PSI/KS drift and benchmarks challenger models.

As detailed in Table 2, each agent operates as an isolated OS process with a dedicated consumer group. The cross-cutting system plane manages heartbeats, runtime discovery, and poison-message isolation via dead-letter queues.

Architectural Rationale: Kafka-Native Actor Model vs. External Stream Engines.

A central design decision in ACIS-X is using a unified Kafka event log rather than introducing external distributed stream-compute clusters (e.g., Apache Flink). While dedicated streaming engines offer distributed state backends, they introduce substantial operational complexity, dual-cluster synchronization overhead, and elevated total cost of ownership (TCO). ACIS-X avoids this by exploiting Kafka partition keying: all business events are strictly keyed by `customer_id`, ensuring per-customer events are routed to the same partition and processed in strict temporal order. This enables low-contention, per-customer ordered in-memory state tracking within individual agent instances without requiring distributed join coordinators. Local state is periodically backed by SQLite or PostgreSQL, while committed Kafka offsets provide deterministic disaster recovery.

4 Methodology

4.1 Three-Tier Signal Fusion

Tier 1 – Behavioural signals. The customer state agent rebuilds metrics from invoices and payments under a per-customer lock. Let $\mathcal{I}(c_i)$ be customer c_i ’s invoice set, I_k an invoice, and $\mathcal{P}$ the payment set. With payment/due timestamps

paid_k , due_k , on-time ratio (OTR) and average delay are:

$$\begin{aligned} \text{OTR}(c_i) &= \frac{|\{I_k : \text{paid}(I_k) \leq \text{due}(I_k)\}|}{|\mathcal{I}(c_i)|}, \\ \text{AvgDelay}(c_i) &= \frac{1}{|\mathcal{P}|} \sum_k \max((\text{paid}_k - \text{due}_k), 0). \end{aligned} \quad (2)$$

Tier 2 – External signals. The external data agent caches financial ratios for 90 days, mapping them to financial risk $\mathcal{E}(c_i) \in [0, 1]$. Litigation scraping scores court exposure over 365 days as $\mathcal{L}(c_i) \in [0, 1]$.

Tier 3 – Fusion. The aggregator agent fuses external signals as:

$$R_{\text{ext}}(c_i) = w_{\text{fin}} \cdot \mathcal{E}(c_i) + w_{\text{lit}} \cdot \mathcal{L}(c_i) \in [0, 1], \quad (3)$$

with baseline weights $w_{\text{fin}} = 0.60$ and $w_{\text{lit}} = 0.40$, treating fundamentals as the primary baseline and litigation as a severe but sparser supplement. (We evaluate sensitivity to these weights in Section 6.2).

4.2 Payment Risk Prediction and Regulatory Guardrails

The payment prediction agent scores each open invoice. Feature vector $X \in \mathbb{R}^d$ encompasses normalised delay, credit utilisation, invoice ratio, late rate, external/litigation terms, and rating penalty. A Random Forest ($M = 50$ trees) yields base probability $r_{ml} = \frac{1}{M} \sum_{m=1}^M h_m(X)$:

$$r_{\text{final}} = \max(0, \min(1, r_{ml} + r_{\text{policy}})). \quad (4)$$

Here r_{policy} represents a deterministic regulatory guardrail: rating penalties (+0.10 for B, +0.20 for C) plus litigation adjustments (+0.15 if $\mathcal{L} > 0.1$, +0.25 if $\mathcal{L} > 0.5$). In enterprise credit governance, rare but catastrophic tail risks (e.g., active insolvency litigation) must not be smoothed out by statistical ensembles. Decoupling statistical r_{ml} from non-negotiable policy constraints r_{policy} guarantees compliance with credit risk management policies.

For auditability, we persist exact SHAP attributions [6] computed in polynomial time via TreeExplainer: $f(X) = E[f(X)] + \sum_{j=1}^d \phi_j$. Policy additives sit outside this decomposition and are logged alongside the SHAP vector for complete mathematical transparency.

4.3 Risk Refinement and Self-Healing Protocol

The risk scoring agent forms the adjusted risk r_{adj} from prediction r_{pred} :

$$r_{\text{adj}} = \max\left(0, \min\left(1, r_{\text{pred}} + 0.85 \cdot \Delta_{\text{behavioural}} + 0.30 \cdot \Delta_{\text{temporal}}\right)\right), \quad (5)$$

where $\Delta_{\text{behavioural}} = 0.40a_{\text{overdue}} + 0.20a_{\text{payment}} + 0.15a_{\text{delay}} + 0.15a_{\text{outstanding}} + 0.10a_{\text{exposure}}$ ranks operational urgency across normalised factors $a_i \in [0, 1]$. Temporal velocity $\Delta_{\text{temporal}} \in [-0.2, 0.3]$ captures trend direction.

Self-Healing Protocol. Agents emit heartbeats every $T_H = 10$ s. Monitoring detects a timeout after 60 s of silence (6 missed beats), triggering an event-driven restart request to resume consumer execution from the last committed offset.

4.4 Autonomous Model Governance and Challenger Benchmarking

To support model-risk governance principles aligned with SR 11-7, Layer 6 introduces the `ModelGovernanceAgent`. It tracks streaming predictions across sliding windows ($W = 200$) and computes the Population Stability Index (PSI):

$$\text{PSI} = \sum_{b=1}^B (P_b - Q_b) \cdot \ln \left(\frac{P_b}{Q_b} \right), \quad (6)$$

where P_b and Q_b are baseline and current empirical quantile proportions across $B = 10$ bins. Values $\text{PSI} \geq 0.10$ trigger automated drift warnings. In parallel, an online challenger model (linear SGD classifier) is continuously trained and benchmarked against the champion Random Forest, publishing periodic audit reports to `acis.system`.

4.5 Jump-Diffusion Synthetic Stress Test Harness

To evaluate ACIS-X across diverse macroeconomic stress cases without exposing confidential commercial ledgers, we implement a jump-diffusion synthetic stress test harness:

$$dZ_t = \kappa(\mu - Z_t)dt + \sigma_Z dW_t + J_t dN_t, \quad (7)$$

where $Z_t \in [0, 1]$ is a latent enterprise stress factor, κ the mean-reversion rate, σ_Z diffusion volatility, and $J_t dN_t$ a Poisson jump process representing liquidity shocks. The harness conditions customer delinquency on Z_t via fat-tailed Pareto delay mixtures across Calm, Stressed, and Crisis regimes. This jump-diffusion harness is used to generate diverse and reproducible stress scenarios for architectural and comparative evaluation, not to establish production-level statistical fidelity.

5 Implementation

The system is implemented in Python 3.11 across ~ 55 files in seven packages: agents, governance, runtime, self-healing, monitoring, registry, and schemas. An entry point pre-creates Kafka topics and spawns agents as isolated OS processes under supervisor control (capped at 5 restarts/2 min). Messages follow a Pydantic-validated envelope (schema v1.1) with correlation IDs; validation failures route to dead-letter queues.

Native Kafka Exactly-Once Semantics (EOS). Transport-layer deduplication is enforced via Kafka idempotent producers with all-in-sync-replica acknowledgements and bounded in-flight requests. Consumers operate with committed read isolation to process solely committed transaction records. When transactional processing is enabled, Kafka transactions provide the primary atomicity guarantee for consume-transform-produce operations via transactional offset commits, ensuring atomicity across topic boundaries. In-memory LRU filters and database primary-key uniqueness constraints are retained downstream as a defense-in-depth mechanism against redundant consumer reconnections or external replay.

6 Experimental Evaluation

6.1 Test Infrastructure and Systems Metrics

The test suite executed on Windows 11 (i7-12700H, 32GB) with Docker Kafka active: **180 passed** (100%), 0 failed, across unit, integration, governance, and fault tests. Table 3 reports key systems metrics.

Table 3. Systems metrics from the automated suite

Metric	Value	Source
Pipeline P95 latency (mocked, $n = 500$)	16 ms (mean 4.6 ms)	performance suite
Collections dedup under 10k repeats	1 action (perfect dedup)	performance suite
Risk-scoring scaling	$3.49\times$ at $N=4$ ($4.93\times$ at $N=6$)	throughput test
Model governance PSI calculation	< 0.10 stable, ≥ 0.25 drift	governance suite
Suite result	180/180 pass (100%)	complete test suite

As shown in Table 3, pipeline P95 latency remains well within sub-50 ms streaming targets, while horizontal scaling demonstrates near-linear throughput increases across dedicated consumer partitions.

6.2 Architectural Ablation and Sensitivity Analysis

To quantify component contributions, we performed an architectural ablation on $n=200$ synthetic customers across 5 random seeds (42, 123, 456, 789, 1024). Table 4 presents Spearman rank correlation (ρ) with ground-truth risk (mean $\pm$ std).

Table 4. Architectural ablation ($n=200$, 5 seeds). ρ = Spearman rank correlation with ground truth (mean $\pm$ std).

Configuration	ρ	$\Delta\rho$	Note
Full ACIS-X (baseline)	0.713 ± 0.024	—	all layers active
–External enrichment	0.711 ± 0.027	-0.002	Tier 2/3 signals ablated
–Behavioural refinement	0.653 ± 0.033	-0.060	raw utilisation only
–SHAP audit layer	(latency)	$+0.9$ ms/event	explainability overhead

Table 4 demonstrates that behavioural refinement provides a substantial ranking lift across the five-seed evaluation ($\Delta\rho = +0.060$, paired $t = 10.47$, $p < 0.001$), while external enrichment’s marginal contribution ($\Delta\rho = +0.002$, $p = 0.27$) is small and not clearly distinguishable from noise at $n = 200$ across 5 seeds. This indicates that real-time behavioural transaction dynamics are the primary operational driver of intraday credit differentiation, with external ratios acting primarily as secondary policy guardrails. SHAP TreeExplainer adds merely ~ 0.9 ms per event.

Weight Sensitivity Analysis. To evaluate the sensitivity of the 0.60/0.40 fusion heuristic, we evaluated weight splits across 50/50, 60/40, and 70/30 ($w_{\text{fin}}/w_{\text{lit}}$) over 5 seeds. Ranking correlation with ground truth remains positive across all configurations ($\rho_{50/50} = 0.761 \pm 0.004$, $\rho_{60/40} = 0.873 \pm 0.004$, $\rho_{70/30} = 0.941 \pm 0.002$), while increasing with financial weight w_{fin} . This upward trend is consistent with the synthetic ground-truth risk function (Eq. 1), which incorporates external financial health alongside behavioural delinquency. Rather

than asserting model-free optimality, this analysis demonstrates that the fusion pipeline responds predictably to signal weighting within the tested synthetic scenarios without requiring complex online reinforcement learning.

We evaluated the self-healing plane by injecting controlled faults into the SelfHealingAgent (Table 5).

Table 5. Fault injection results (mocked Kafka, single-node decision latency).

Scenario	Metric	Value	Assertion
Agent hard crash (TIMEOUT)	decision latency	0.26 ms	< 100 ms
Latency spike (5 s, CPU 95%)	decision latency	0.10 ms	< 100 ms
Cascading 5-agent failure	restarts/agent	1	exactly 1 (storm prevention)
100-cycle endurance	success rate	100%	100/100
100-cycle endurance	P95 decision latency	0.23 ms	< 50 ms

Table 5 shows that self-healing decision latency is sub-millisecond (0.26 ms for hard crashes under single-node conditions). The cooldown mechanism provides storm prevention (1 restart per agent during 5-agent cascading failure) with 100% success across 100 endurance cycles (P95 decision latency 0.23 ms).

6.3 Predictive Comparison and Methodological Nuance

To evaluate risk scoring capabilities, we benchmark ACIS-X against naive heuristics and conventional supervised learners on a synthetic hold-out dataset (Table 6, $n = 500$, 70/30 split, mean $\pm$ std across 5 seeds: 42, 123, 456, 789, 1024).

Table 6. Predictive benchmark on synthetic hold-out ($n = 500$, 5 seeds, mean $\pm$ std). Supervised models are trained directly on generator ground-truth labels; ACIS-X operates label-free.

Method	F1	AUC
Naive utilisation threshold	0.730 ± 0.034	—
ACIS-X streaming refinement (label-free)	0.758 ± 0.016	0.862 ± 0.025
Logistic Regression (basic / enriched, supervised)	0.929 ± 0.027 / 0.967 ± 0.017	0.988 ± 0.006 / 0.997 ± 0.002
Random Forest (basic / enriched, supervised)	0.889 ± 0.024 / 0.895 ± 0.029	0.970 ± 0.013 / 0.975 ± 0.011
Gradient Boosting (basic / enriched, supervised)	0.869 ± 0.032 / 0.887 ± 0.039	0.965 ± 0.012 / 0.970 ± 0.013

Table 6 illustrates an essential architectural distinction: conventional supervised models (Logistic Regression, Random Forest, GBDT) achieve high metrics (AUC up to 0.997) because they are trained with full supervision directly on the synthetic generator’s mathematical labelling function. In contrast, ACIS-X operates as an online, label-free streaming framework that requires no historical default training labels. In this label-free setting, ACIS-X multi-signal refinement yields a modest mean F1 improvement over the naive heuristic rule (0.758 ± 0.016 vs 0.730 ± 0.034 , winning in 4/5 seed trials, paired $t = 1.42, p = 0.23$) while critically providing continuous probability ranking and temporal velocity (AUC 0.862 ± 0.025) that single-threshold heuristics cannot generate. Supervised models confirm that external features provide strong discriminative signal when explicit default labels are available.

7 Discussion and Limitations

The ablation (Table 4) confirms behavioural refinement is the primary driver of rank quality ($\Delta\rho = 0.060, p < 0.001$), with SHAP explainability adding under 1 ms per event. Fault injection (Table 5) demonstrates sub-millisecond self-healing decision latency under single-node conditions with zero restart storms. Note that this metric measures agent decision and restart-dispatch latency; in a distributed production deployment, end-to-end recovery would also be governed by Kafka consumer group rebalance timers (typically 3–10 s). The governance plane demonstrates that continuous statistical monitoring (PSI and KS testing) can operate online alongside event-driven pipelines without degrading streaming throughput.

Limitations and Domain Context. We explicitly acknowledge four core limitations: (1) *Synthetic vs. real data*: synthetic stochastic generation models stylized delinquency dynamics but cannot fully replicate non-linear macroeconomic contagions or strategic invoice deferrals; (2) *Evaluation scope*: results represent architectural and systems validation (latency, scaling, self-healing, idempotency, and fusion) under controlled conditions; (3) *Production readiness*: deployment requires empirical backtesting on enterprise ERP ledgers and multi-datacenter disaster recovery drills; and (4) *Sample size of multi-seed trials*: paired significance tests use $n = 5$ random seeds; while ranking trends are consistent across seeds, larger seed sets ($n \geq 30$) will provide greater statistical power. Furthermore, the 60-second heartbeat timeout is tailored for B2B trade credit (30–60 day terms), where a 60-second window provides near-instantaneous intraday response compared to 24–168 hour batch cycles, while suppressing failover flapping from transient GC pauses.

8 Conclusion

ACIS-X is a Kafka reactive multi-agent credit pipeline featuring three-tier fusion, native EOS transport guarantees, autonomous model governance, SHAP audit persistence, and heartbeat-driven recovery. On the automated test suite, all 180 tests passed; pipeline P95 latency reached 16 ms and scaling reached $3.49\times$ at 4 workers. Architectural ablation confirmed behavioural refinement delivers the largest rank lift ($\Delta\rho = 0.060$) with < 1 ms SHAP overhead, while Sensitivity analysis showed a predictable increase in ranking correlation with financial-signal weight within the tested synthetic scenarios. Fault injection demonstrated sub-millisecond self-healing decision latency under controlled conditions with 100% endurance success. Future work includes empirical validation on enterprise ERP ledgers and multi-cluster disaster recovery drills.

Acknowledgement

The authors thank the faculty and laboratory infrastructure of the Department of Computer Science and Engineering, AMC Engineering College, Bengaluru, for supporting this research.

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